

	$\mathcal{A}_{1^+ \alpha \beta}^{\#1}$	$\mathcal{A}_{1^+ \alpha \beta}^{\#2}$	$f_{1^+ \alpha \beta}^{\#1}$	$\mathcal{A}_{1^- \alpha}^{\#1}$	$\mathcal{A}_{1^- \alpha}^{\#2}$	$f_{1^- \alpha}^{\#1}$	$f_{1^- \alpha}^{\#2}$
$\mathcal{A}_{1^+}^{\#1} + \beta$	$\frac{1}{4} (12 \beta_1 - 10 \beta_2 + 2 (\alpha_2 - \alpha_3 + 4 \alpha_4 - 4 \alpha_6) k^2 + (\mathcal{M}_{\beta_1}^2))$	$\frac{4 \beta_1 - 6 \beta_2 + (\mathcal{M}_{\beta_1}^2)}{2 \sqrt{2}}$	$\frac{i k (4 \beta_1 - 6 \beta_2 + (\mathcal{M}_{\beta_1}^2))}{2 \sqrt{2}}$	0	0	0	0
$\mathcal{A}_{1^+}^{\#2} + \alpha \beta$	$\frac{4 \beta_1 - 6 \beta_2 + (\mathcal{M}_{\beta_1}^2)}{2 \sqrt{2}}$	$2 \beta_1 - \beta_2$	$i (2 \beta_1 - \beta_2) k$	0	0	0	0
$f_{1^+}^{\#1} + \alpha \beta$	$-\frac{i k (4 \beta_1 - 6 \beta_2 + (\mathcal{M}_{\beta_1}^2))}{2 \sqrt{2}}$	$-i (2 \beta_1 - \beta_2) k$	$(2 \beta_1 - \beta_2) k^2$	0	0	0	0
$\mathcal{A}_{1^+}^{\#1} + \alpha$	0	0	0	$\beta_1 + \frac{\beta_2}{2} + \beta_3 + \frac{(\mathcal{M}_{\beta_1}^2)}{4} + \frac{1}{72} (k + 6 k \theta)^2 \xi$	$-\frac{72 \beta_3 + 36 (\mathcal{M}_{\beta_1}^2) + k^2 (1 + 6 \theta) \xi}{72 \sqrt{2}}$	0	$-\frac{1}{72} i k (72 \beta_3 + 36 (\mathcal{M}_{\beta_1}^2) + k^2 (1 + 6 \theta) \xi)$
$\mathcal{A}_{1^+}^{\#2} + \alpha$	0	0	0	$-\frac{72 \beta_3 + 36 (\mathcal{M}_{\beta_1}^2) + k^2 (1 + 6 \theta) \xi}{72 \sqrt{2}}$	$\beta_1 + \frac{\beta_2 + \beta_3}{2} + \frac{k^2 \xi}{144}$	0	$\frac{i k (72 (2 \beta_1 + \beta_2 + \beta_3) + k^2 \xi)}{72 \sqrt{2}}$
$f_{1^+}^{\#1} + \alpha$	0	0	0	0	0	0	0
$f_{1^+}^{\#2} + \alpha$	0	0	0	$\frac{1}{72} i k (72 \beta_3 + 36 (\mathcal{M}_{\beta_1}^2) + k^2 (1 + 6 \theta) \xi)$	$-\frac{i k (72 (2 \beta_1 + \beta_2 + \beta_3) + k^2 \xi)}{72 \sqrt{2}}$	0	$(2 \beta_1 + \beta_2 + \beta_3) k^2 + \frac{k^4 \xi}{72}$

	$\mathcal{A}_{2^+ \alpha \beta}^{\#1}$	$\mathcal{F}_{2^+ \alpha \beta}^{\#1}$	$\mathcal{A}_{2^- \alpha \beta \chi}^{\#2}$
$\mathcal{A}_{2^+}^{\#1} \uparrow^{\alpha \beta}$	$\frac{1}{4} (4 \beta_1 + 2 \beta_2 - (\mathcal{M}_{\text{Pl}}^2) + 2 k^2 (-3 \alpha_2 + \alpha_3 - 4 \alpha_4 + 4 \alpha_6 + 2 \theta^2 \xi))$	$-\frac{i k (4 \beta_1 + 2 \beta_2 - (\mathcal{M}_{\text{Pl}}^2))}{2 \sqrt{2}}$	0
$\mathcal{F}_{2^+}^{\#1} \uparrow^{\alpha \beta}$	$\frac{i k (4 \beta_1 + 2 \beta_2 - (\mathcal{M}_{\text{Pl}}^2))}{2 \sqrt{2}}$	$(2 \beta_1 + \beta_2) k^2$	0
$\mathcal{A}_{2^+}^{\#1} \uparrow^{\alpha \beta \chi}$	0	0	$\frac{1}{4} (4 \beta_1 + 2 \beta_2 - (\mathcal{M}_{\text{Pl}}^2) + 2 k^2 (-2 \alpha_2 + \theta^2 \xi))$

	$\sigma_{2^+ \alpha \beta}^{\#1}$	$\tau_{2^+ \alpha \beta}^{\#1}$	$\sigma_{2^+ \alpha \beta X}^{\#2}$
$\sigma_{2^+}^{\#1} + \alpha \beta$	$\frac{(\beta_1 + \beta_2) k^2}{\text{Det}(2^+)}$	$\frac{i k (4 \beta_1 + 2 \beta_2 \cdot (M_{\beta_1}^2))}{2 \sqrt{2} \text{Det}(2^+)}$	0
$\tau_{2^+}^{\#1} + \alpha \beta$	$\frac{i k (4 \beta_1 + 2 \beta_2 \cdot (M_{\beta_1}^2))}{2 \sqrt{2} \text{Det}(2^+)}$	$\frac{4 \beta_1 + 2 \beta_2 \cdot (M_{\beta_1}^2) + 2 k^2 \cdot (-3 \alpha_2 + \alpha_3 - 4 \alpha_4 + 4 \alpha_6 + 2 \theta^2 \xi)}{4 \text{Det}(2^+)}$	0
$\sigma_{2^+}^{\#2} + \alpha \beta X$	0	0	$\frac{1}{\text{Det}(2^+)}$

$$\text{Det}(1^-) \rightarrow \frac{1}{192} (1 + 2 k^2) (24 (4 \beta_1 + 2 \beta_2 - (\mathcal{M}_{\text{pl}}^2)) (2 \beta_1 + \beta_2 + 3 \beta_3 + (\mathcal{M}_{\text{pl}}^2)) + k^2 (4 \beta_1 + 2 \beta_2 - (\mathcal{M}_{\text{pl}}^2) + 8 (4 \beta_1 + 2 \beta_2 - (\mathcal{M}_{\text{pl}}^2)) \theta + 48 (2 \beta_1 + \beta_2 + \beta_3) \theta^2) \xi)$$

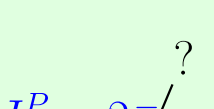
$$\begin{aligned} \text{Det}(0^+) \rightarrow \frac{1}{2} k^2 \\ (-8 \alpha_4 \beta_1 k^2 + 8 \alpha_6 \beta_1 k^2 - 4 \alpha_4 \beta_2 k^2 + 4 \alpha_6 \beta_2 k^2 - 12 \alpha_4 \beta_3 k^2 + 12 \alpha_6 \beta_3 k^2 + \\ 12 \alpha_1 (2 \beta_1 + \beta_2 + 3 \beta_3) k^2 + 4 \alpha_3 (2 \beta_1 + \beta_2 + 3 \beta_3) k^2 - 2 \beta_1 (M_{\beta_1}^2) - \\ \beta_2 (M_{\beta_1}^2) - 3 \beta_3 (M_{\beta_1}^2) - (M_{\beta_1}^2)^2 + 2 (2 \beta_1 + \beta_2 + 3 \beta_3) k^2 \theta^2 \xi) \end{aligned}$$

A diagram showing two vertices connected by a wavy line. The wavy line is labeled k, μ with an arrow pointing from the left vertex to the right vertex. The vertices are labeled $j^P = 1^-$ on the right. Each vertex has two external lines, one of which is labeled with a question mark $?$.

Diagram illustrating a bubble diagram with two vertices (circles) and four external legs (lines ending in question marks). A dashed red line connects the two vertices, labeled with a red arrow and the text $j^P = 0_+$ and k^μ .

Diagram illustrating a box diagram with two internal vertices (circles) and two external legs (lines with question marks). A wavy line connects the vertices, labeled with $j^P = 2+$ and k, μ .

A diagram showing two vertices (pink circles) connected by a horizontal line. A green arrow labeled k^μ points from the left vertex to the right vertex. Each vertex has two external legs, all four of which are marked with a question mark.

		Massive particle
Pole residue:	$\frac{2}{2 a_2 - 2 \theta^2 \xi} > 0$	
Polarisations:	5	
Square mass:	$\frac{4 \beta_1 + 2 \beta_2 \cdot (M_0^2)}{4 a_2 - 2 \theta^2 \xi} > 0$	
Spin:	2	
Parity:	Odd	

Unitarity conditions
(Timeout after 20 seconds)